

Devasurya J Menon

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CAREER OBJECTIVE

Final-year Computer Science student targeting AI Engineer roles, with deployed projects across LLM agents, applied machine learning, and backend infrastructure. Interested in how systems work in the real world, and in building things that are genuinely useful.

EDUCATION

Jain (Deemed-to-be) University — Bangalore, Karnataka, India

Aug 2023 – May 2027

Bachelor of Technology (B.Tech), Computer Science & Engineering · CGPA: 8.7 / 10

PROJECT EXPERIENCE

TraceFlow — Multi-Agent Root Cause Analysis Engine [[GitHub](#)]

Jun 2026 – Jul 2026

Python, FastAPI, PostgreSQL, ChromaDB, Gemini API, Docker, pytest, GitHub Actions, React

- Designed an asynchronous diagnostic microservice for factory fault telemetry: FastAPI logs the incident and responds immediately, then three LLM agents — Investigator, Diagnostician, Verifier — reason over PostgreSQL fault history and a ChromaDB index of 8 manuals, checking every claim against retrieved evidence.
- Measured quality on 13 gold-labeled incidents: retrieval recall at rank 3 of 12 out of 13, and 36 out of 39 on a grounding rubric, with token, cost, and latency logged per incident.
- Secured the service with JWT authentication, scoped SQL and vector filters enforced under test, call-budget caps, and a prompt-injection defense, with continuous integration running the test suite on every push.

Drifty — Prompt Refinement and Drift Monitoring Platform [[GitHub](#)]

Jul 2026 – Aug 2026

Python, FastAPI, Redis, PostgreSQL, React, TypeScript, Docker, Groq and Gemini APIs

- Built a platform that turns a plain-English goal into a production prompt: three models draft candidates in different styles and rank each other's anonymized attempts with order randomized to cancel judge position bias, then the top two run against the user's own inputs so the winner is chosen on output, not on how it reads.
- Hand-rolled the background job queue on two Redis lists with blocking-pop priority lanes and atomic fan-in counters instead of Celery or RQ — roughly 90 lines, with partial-failure handling.
- Scheduled a daily job that re-runs every registered prompt against four pluggable checks, surfacing drift as a rolling pass-rate window on a React dashboard with latency, token, and cost metrics.

Explainable Eye Disease Detection [[Live Demo](#) · [GitHub](#)]

Apr 2026

PyTorch, ResNet18, Swin Transformer, Grad-CAM, Gradio, Hugging Face Spaces

- Trained and compared a ResNet18 convolutional network and a Swin-Tiny vision transformer on 4,217 retinal images across 4 disease classes, reaching 91.12% and 94.08% validation accuracy against a 25% baseline.
- Added Grad-CAM heatmaps and clinical alignment logic that maps each disease to its expected anatomical region, flagging predictions that are correct for the wrong reason, and deployed both models with the explainability interface live on Hugging Face Spaces via Gradio.

SKILLS

- **Programming Languages:** Python, JavaScript, Java, SQL
- **Machine Learning and AI:** PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, Grad-CAM, SHAP, Retrieval-Augmented Generation (RAG), multi-agent LLM pipelines, prompt engineering, model evaluation
- **Backend and Data:** FastAPI, Flask, REST API design, asynchronous Python, PostgreSQL, MySQL, MongoDB, Redis, ChromaDB, Pandas
- **Frontend, DevOps and Tooling:** React, Vite, Streamlit, Gradio, Docker, GitHub Actions, CI/CD, Git, Linux, pytest, Hugging Face Spaces
- **Core Concepts:** Data Structures and Algorithms, Object-Oriented Programming

ACHIEVEMENTS

- **Research Publication:** Co-authored a paper accepted at a Springer–IEEE conference on comparative analysis of machine learning models for diabetes prediction, benchmarking 7 models including Random Forest, XGBoost, and CatBoost to a peak accuracy of 70.13%.
- **Published Claude Agent Skill:** Authored and open-sourced *context-handoff* ([GitHub](#)), an MIT-licensed skill that keeps architecture and task-log documents current so a new AI session resumes a project with no context loss. Submitted to the official Claude marketplace.
- **Japanese Language (JLPT N5):** Reached N5 proficiency through independent self-study alongside full-time coursework.